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2027 CYCLE

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CUET UG

Physics (322)

Atoms and Nuclei

Official Syllabus Unit 8 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 8)	Key Exam Focus & Skills
Alpha Scattering & Rutherford Atom	Alpha-particle scattering experiment; Rutherford's model of atom.	Geiger-Marsden experiment; impact parameter b; distance of closest approach $r_0 = \frac{2Ze^2}{4\pi\varepsilon_0 K_\alpha}$; atomic instability and continuous spectrum failure.
Bohr Hydrogen Model & Orbitals	Bohr model of hydrogen atom, Expression for radius of an orbit, velocity and energy of electron in an orbit.	Bohr postulates; orbital radius $r_n = 0.529\left(\frac{n^2}{Z}\right)$ Å; speed $v_n = 2.18 \times 10^6\left(\frac{Z}{n}\right)$ m/s; energy scaling $E_n = -13.6\left(\frac{Z^2}{n^2}\right)$ eV; relations $K = -E_n$, $U = 2E_n$.
Hydrogen Spectral Series	Hydrogen line spectra (qualitative treatment only).	Rydberg formula $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$; Lyman (UV), Balmer (Visible), Paschen/Brackett/Pfund (IR); series limits and total emitted lines $N = \frac{n(n-1)}{2}$.
Nuclear Size, Density & Forces	Composition and size of nucleus, atomic number, atomic mass number, isotopes, isobars and isotones, nuclear force.	Nuclear radius $R = R_0 A^{\frac{1}{3}}$; constant nuclear density $\rho \approx 2.3 \times 10^{17}$ kg/m^3; strong force properties: short-range, charge-independent, saturated, repulsive core.
Mass Defect & Binding Energy	Mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number.	Einstein equivalence 1 u = 931.5 MeV; mass defect Δm; binding energy curve with peak at Iron-56 (8.75 MeV/nucleon); plateau at 8.5 MeV; drop at high/low A.
Nuclear Fission & Fusion	Nuclear fission, nuclear fusion.	Fission of Uranium-235 by thermal neutrons ($\sim$ 200 MeV, 2–3 neutrons); thermonuclear fusion (proton-proton chain in stars,

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		26.7 MeV); energy density comparison (MeV/amu).

EXAM PATTERN

CUET UG Exam Architecture: Unit 8 carries substantial weightage in Section II Physics (typically 4 to 5 direct MCQs). The questions rigorously test numerical scaling ($r \propto n^2$, $v \propto \frac{1}{n}$, $E \propto \frac{1}{n^2}$), energy differences in hydrogen transitions, nuclear radius ratio ($\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}}$), constant nuclear density, and Q-value calculations in fission and fusion.

Concept Module 1: Alpha-Particle Scattering & Rutherford's Nuclear Model

The Geiger-Marsden Experiment (1911)

Under the direction of Ernest Rutherford, H. Geiger and E. Marsden directed a collimated beam of 5.5 MeV alpha particles ($\alpha = {}_2^4\mathrm{He}^{2+}$) from a bismuth radioactive source (${}_{83}^{214}\mathrm{Bi}$) onto a thin gold foil (thickness $t \approx 2.1 \times 10^{-7}$ m ≈ 400 atoms thick). The scattered alpha particles struck a zinc sulfide (ZnS) fluorescent screen, producing tiny scintillations observed through a rotatable microscope.

IMPORTANT CONCEPT

Three Landmark Experimental Observations:

1. **Vast Majority Undeflected:** About 99.86% of the incident alpha particles passed straight through the gold foil with little or no deflection ($\theta < 1°$).
2. **Occasional Deflection:** Only about 1 in 8000 alpha particles was deflected through large angles ($\theta > 90°$).
3. **Rare Rebounds:** Extremely rarely, an alpha particle was deflected backwards ($\theta \approx 180°$).

Rutherford's Conclusions & Nuclear Atom

- **Atom is Mostly Empty Space:** Since nearly all alpha particles passed unhindered, the bulk volume of an atom is vacant space.
- **Concentrated Positive Core (Nucleus):** To deflect a massive, doubly-charged alpha particle by 180°, the entire positive charge of the atom and almost its entire mass must be densely concentrated in an exceedingly small volume at the centre, termed the **Nucleus**.
- **Planetary Electron Cloud:** Negatively charged electrons revolve around the central positive nucleus in circular orbits, analogous to planets revolving around the Sun.

Distance of Closest Approach (r_0)

In a head-on collision (scattering angle $\theta = 180°$), an alpha particle approaches the target nucleus along the central line of centres. As it approaches, its initial kinetic energy K_α is continuously converted into electrostatic potential energy until it momentarily stops at distance r_0 and recoils backward:

$$K_\alpha = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0} = \frac{2Ze^2}{4\pi\varepsilon_0 r_0}$$

Solving for the distance of closest approach r_0:

$$r_0 = \frac{2Ze^2}{4\pi\varepsilon_0 K_\alpha}$$

For a 5.5 MeV alpha particle striking Gold ($Z = 79$):

$$r_0 = \frac{9\times10^9 \times 2 \times 79 \times (1.6\times10^{-19})^2}{5.5\times1.6\times10^{-13}\ \text{J}} \approx 4.13\times10^{-14}\ \text{m} = 41.3\ \text{fm}$$

This established that the radius of a gold nucleus must be strictly less than 41.3 fm (in reality, $R_{\text{Au}} \approx 7$ fm).

Impact Parameter (b) & Scattering Distribution

The **impact parameter** (b) is the perpendicular distance of the initial velocity vector of the alpha particle from the central line of the target nucleus:

$$b = \frac{Ze^2\cot\left(\frac{\theta}{2}\right)}{4\pi\varepsilon_0 K_\alpha}$$

- If b is large, electrostatic repulsion is weak, and the deflection angle is small ($\theta \approx 0°$).
- If $b = 0$ (head-on collision), $\cot\left(\frac{\theta}{2}\right) = 0 \Longrightarrow \theta = 180°$.
- **Scattering Angular Distribution Law:** The number of alpha particles $N(\theta)$ scattered per unit area at angle θ satisfies:

$$N(\theta) \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$

Limitations of Rutherford's Model

1. **Instability of the Atom:** According to classical Maxwellian electrodynamics, an electron in a circular orbit experiences centripetal acceleration ($a = \frac{v^2}{r}$) and must continuously radiate electromagnetic energy. Its orbital radius must decrease continuously, spiraling into the nucleus in approximately 10^{-8} seconds.
2. **Inability to Explain Discrete Line Spectra:** A spiraling electron would radiate a continuous spectrum of all frequencies, contradicting the discrete line spectra observed experimentally in gas discharges.

DISTANCE OF CLOSEST APPROACH FOR ALPHA PARTICLE

Problem: An alpha particle of kinetic energy 7.7 MeV is fired head-on at a gold nucleus ($Z = 79$). Find the distance of closest approach ($\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ N m^{2}C^{-2}).

Solution: At closest approach, $K_\alpha = \frac{2Ze^2}{4\pi\varepsilon_0 r_0} \Longrightarrow r_0 = \frac{2Ze^2}{4\pi\varepsilon_0 K_\alpha}$. With $K_\alpha = 7.7 \times 1.6 \times 10^{-13}$ J $= 1.232 \times 10^{-12}$ J:

$$r_0 = \frac{9 \times 10^9 \times 2 \times 79 \times \left(1.6 \times 10^{-19}\right)^2}{1.232 \times 10^{-12}} \approx 2.95 \times 10^{-14} \text{ m} = 29.5 \text{ fm}$$

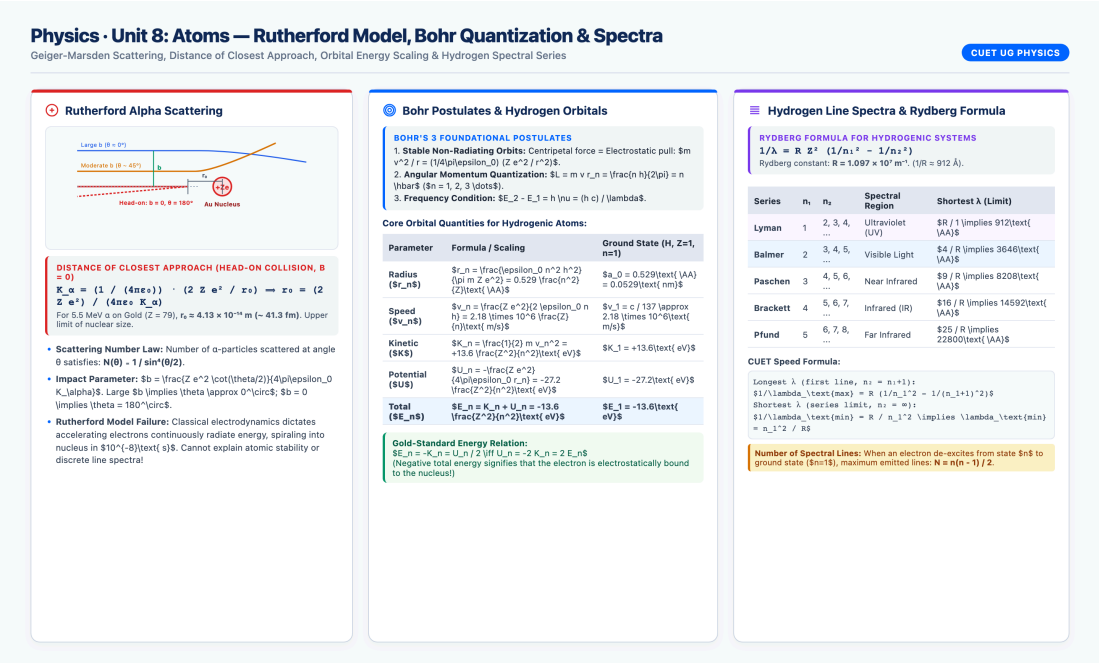
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Concept Module 2: Bohr's Model of the Hydrogen Atom

Bohr's Three Foundational Postulates (1913)

To overcome the limitations of Rutherford's model, Niels Bohr incorporated Planck's quantum theory into the planetary atom:

1. **Postulate 1 (Stationary Orbits):** An electron revolves around the nucleus only in certain discrete, non-radiating circular orbits called **stationary states**: $\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2}$.
2. **Postulate 2 (Quantization of Angular Momentum):** $L = mvr_n = \frac{nh}{2\pi} = n\hbar, \quad n = 1, 2, 3, \ldots$
3. **Postulate 3 (Frequency Condition):** Radiation is emitted/absorbed on quantum jumps: $E_2 - E_1 = h\nu = \frac{hc}{\lambda}$.



Derivation of Orbital Radius (r_n)

From Postulate 2, the speed in orbit n is:

$$v_n = \frac{nh}{2\pi m r_n}$$

Substituting v_n into the centripetal equation:

$$\frac{m}{r_n}\left(\frac{nh}{2\pi m r_n}\right)^2 = \frac{Ze^2}{4\pi\varepsilon_0 r_n^2} \Longrightarrow \frac{n^2h^2}{4\pi^2 m r_n^3} = \frac{Ze^2}{4\pi\varepsilon_0 r_n^2}$$

Solving for radius r_n:

$$r_n = \frac{\varepsilon_0 n^2 h^2}{\pi m Z e^2}$$

For hydrogen ($Z = 1$):

$$r_n = n^2 \left(\frac{\varepsilon_0 h^2}{\pi m e^2} \right) = n^2 a_0$$

where $a_0 = 0.529 \times 10^{-10}$ m $= 0.529$ Å is the **Bohr radius** (ground state radius, $n = 1$). In general:

$$r_n \propto \frac{n^2}{Z}$$

Orbital Speed (v_n)

Substituting r_n into the angular momentum equation:

$$v_n = \frac{Ze^2}{2\varepsilon_0 nh} = \left(\frac{c}{137} \right) \left(\frac{Z}{n} \right) \approx (2.18 \times 10^6) \left(\frac{Z}{n} \right) \text{ m/s}$$

where $\alpha = \frac{e^2}{2\varepsilon_0 hc} \approx \frac{1}{137}$ is the **fine-structure constant**. Notice that:

$$v_n \propto \frac{Z}{n}$$

Kinetic, Potential, and Total Energy in the n-th Orbit

1. **Kinetic Energy (K_n):**

$$K_n = \frac{1}{2} m v_n^2 = \frac{1}{8\pi\varepsilon_0} \frac{Ze^2}{r_n} = +13.6 \left(\frac{Z^2}{n^2} \right) \text{ eV}$$

2. **Potential Energy (U_n):**

$$U_n = -\frac{1}{4\pi\varepsilon_0} \frac{Ze^2}{r_n} = -27.2 \left(\frac{Z^2}{n^2} \right) \text{ eV}$$

3. **Total Mechanical Energy (E_n):**

$$E_n = K_n + U_n = -\frac{1}{8\pi\varepsilon_0} \frac{Ze^2}{r_n} = -13.6 \left(\frac{Z^2}{n^2} \right) \text{ eV}$$

IMPORTANT CONCEPT

Universal Energy Interrelations:

$$E_n = -K_n = \frac{U_n}{2}$$
$$U_n = -2K_n = 2E_n$$

- The total energy E_n is strictly negative, confirming that the electron is electrostatically bound to the nucleus.
- As $n \to \infty$, $E_\infty \to 0$, meaning the electron is freed from the atom.

- **Ionization Energy:** The energy required to liberate an electron from the ground state ($n = 1$) to infinity ($E_1 = -13.6$ eV $\Longrightarrow E_{\text{ion}} = +13.6$ eV).

ORBITAL CALCULATIONS FOR SINGLY IONIZED HELIUM

Problem: For a singly ionized helium atom ($^4_2\text{He}^+$, $Z = 2$): (a) Find the radius of the first excited state ($n = 2$). (b) Find the total energy, kinetic energy, and potential energy in this state.

Solution: (a) Radius of $n = 2$ state:

$$r_2 = 0.529\left(\frac{n^2}{Z}\right)\ \text{Å} = 0.529\left(\frac{2^2}{2}\right) = 0.529 \times 2 = 1.058\ \text{Å}$$

(b) Total energy in $n = 2$:

$$E_2 = -13.6\left(\frac{Z^2}{n^2}\right)\ \text{eV} = -13.6\left(\frac{2^2}{2^2}\right) = -13.6\ \text{eV}$$

Kinetic energy: $K_2 = -E_2 = +13.6$ eV. Potential energy: $U_2 = 2E_2 = -27.2$ eV.

Concept Module 3: Hydrogen Spectral Series & Rydberg Transitions

Derivation of the Rydberg Formula

When an electron transitions from an initial higher stationary state of principal quantum number n_2 to a final lower state n_1 ($n_2 > n_1$), the emitted photon energy is:

$$\Delta E = E_{n_2} - E_{n_1} = h\nu = \frac{hc}{\lambda}$$

Substituting the total energy expression $E_n = -\frac{mZ^2e^4}{8\varepsilon_0^2h^2n^2}$:

$$\frac{1}{\lambda} = \frac{\Delta E}{hc} = \frac{me^4}{8\varepsilon_0^2h^3c}Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Defining the **Rydberg Constant** R:

$$R = \frac{me^4}{8\varepsilon_0^2h^3c} \approx 1.09737 \times 10^7\ \text{m}^{-1}$$

The wave number ($\bar{\nu} = \frac{1}{\lambda}$) is given by the master **Rydberg Equation**:

$$\bar{\nu} = \frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

For hydrogen ($Z = 1$):

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

The Five Spectral Series of Hydrogen

Series	n_1	n_2	Spectral Region	Wavelength Range ($\lambda_{max} \leftrightarrow \lambda_{min}$)
Lyman	1	2, 3, 4, ...	Ultraviolet (UV)	$\lambda_{max} = \frac{4}{3R} \approx 1216$ Å $\lambda_{min} = \frac{1}{R} \approx 912$ Å
Balmer	2	3, 4, 5, ...	Visible Spectrum	$\lambda_{max} = \frac{36}{5R} \approx 6563$ Å (H_α, red) $\lambda_{min} = \frac{4}{R} \approx 3646$ Å
Paschen	3	4, 5, 6, ...	Near Infrared (IR)	$\lambda_{max} = \frac{144}{7R} \approx 18750$ Å $\lambda_{min} = \frac{9}{R} \approx 8208$ Å
Brackett	4	5, 6, 7, ...	Infrared (IR)	$\lambda_{max} = \frac{400}{9R} \approx 40510$ Å $\lambda_{min} = \frac{16}{R} \approx 14592$ Å
Pfund	5	6, 7, 8, ...	Far Infrared (IR)	$\lambda_{max} = \frac{900}{11R} \approx 74580$ Å $\lambda_{min} = \frac{25}{R} \approx 22800$ Å

HIGH-YIELD TIP

CUET High-Yield Shortcut Formulas:

- **Longest Wavelength (λ_{max}):** Occurs for the transition from the immediate next level ($n_2 = n_1 + 1$). Smallest energy jump ΔE_{min}.
- **Shortest Wavelength (Series Limit, λ_{min}):** Occurs for transition from infinity ($n_2 = \infty$). Largest energy jump ΔE_{max}:

$$\frac{1}{\lambda_{min}} = \frac{R}{n_1^2} \Longrightarrow \lambda_{min} = \frac{n_1^2}{R}$$

- **Ratio of $\frac{\lambda_{max}}{\lambda_{min}}$ for Any Series:**

$$\frac{\lambda_{max}}{\lambda_{min}} = \frac{(n_1+1)^2}{(n_1+1)^2 - n_1^2} = \frac{(n_1+1)^2}{2n_1+1}$$

For Lyman ($n_1 = 1$): $\frac{\lambda_{max}}{\lambda_{min}} = \frac{4}{3}$.
For Balmer ($n_1 = 2$): $\frac{\lambda_{max}}{\lambda_{min}} = \frac{9}{5} = 1.8$.

Total Number of Emitted Spectral Lines

When an electron in an excited state of principal quantum number n cascades down to the ground state ($n = 1$), the total number of distinct spectral lines possible is given by combinations:

$$N = C(n, 2) = \frac{n(n-1)}{2}$$

For example, for an electron excited to $n = 4$, the number of lines is $\frac{4\times3}{2} = 6$ lines.

WAVELENGTH RATIO IN BALMER SERIES

Problem: Calculate the ratio of the longest wavelength to the shortest wavelength in the Balmer series of hydrogen.

Solution: For Balmer series, $n_1 = 2$.

- Longest wavelength ($\lambda_{\max}$) corresponds to transition from $n_2 = 3$:

$$\frac{1}{\lambda_{\max}} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36} \Longrightarrow \lambda_{\max} = \frac{36}{5R}$$

- Shortest wavelength ($\lambda_{\min}$, series limit) corresponds to $n_2 = \infty$:

$$\frac{1}{\lambda_{\min}} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R}{4} \Longrightarrow \lambda_{\min} = \frac{4}{R}$$

- Taking the ratio:

$$\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\frac{36}{5R}}{\frac{4}{R}} = \frac{36}{20} = \frac{9}{5} = 1.8$$

Concept Module 4: Nuclear Structure, Size & Nuclear Forces

Nuclear Composition: Protons and Neutrons

The atomic nucleus consists of elementary constituents collectively termed **nucleons**:

- **Protons (p):** Positively charged particles ($q = +e = +1.602 \times 10^{-19}$ C, $m_p = 1.007276$ u $= 1.6726 \times 10^{-27}$ kg).
- **Neutrons (n):** Discovered by James Chadwick (1932) by bombarding Beryllium with alpha particles. Electrically neutral ($q = 0$, $m_n = 1.008665$ u $= 1.6749 \times 10^{-27}$ kg).
- **Atomic Number (Z):** Number of protons in the nucleus (determines chemical element identity and nuclear charge $+Ze$).
- **Mass Number (A):** Total number of nucleons ($A = Z + N$, where N is neutron number).
- **Symbol:** Standard nuclear notation is ${}^{A}_{Z}\mathrm{X}$.

Nuclear Radius Law & Constant Nuclear Density

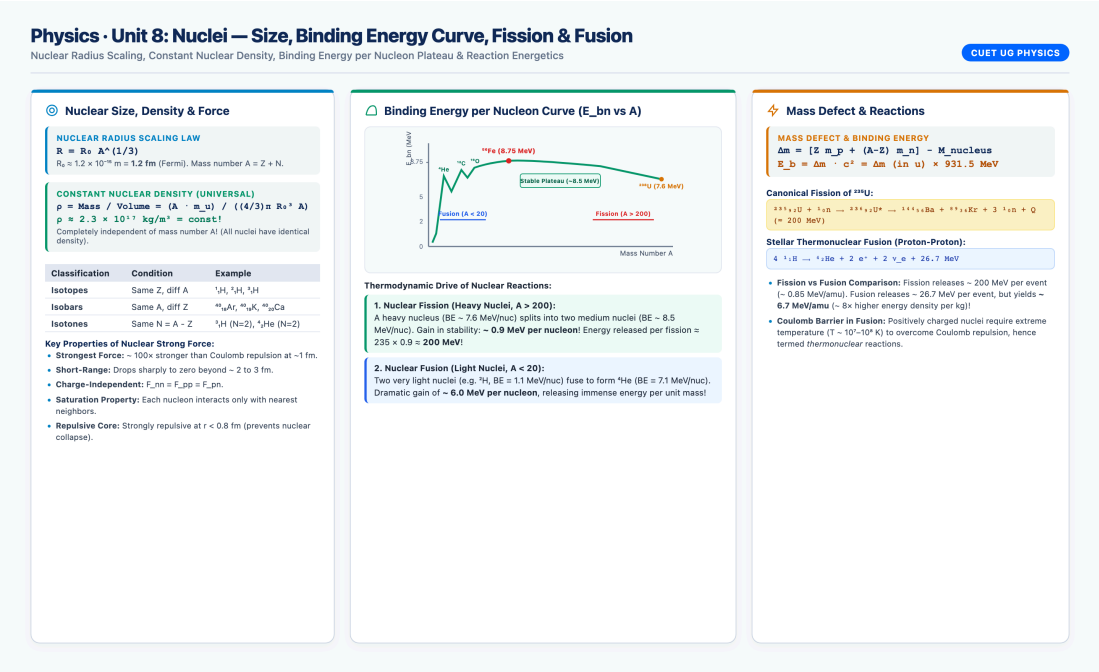
Rutherford's scattering experiments, later refined by high-energy electron scattering measurements, showed that the volume of a nucleus is directly proportional to its mass number A:

$$V = \frac{4}{3}\pi R^3 \propto A \Longrightarrow R \propto A^{\frac{1}{3}}$$

The nuclear radius R follows the empirical formula:

$$R = R_0 A^{\frac{1}{3}}$$

where $R_0 = (1.20 \pm 0.1) \times 10^{-15}$ m ≈ 1.2 fm.



Proof that Nuclear Density is Independent of Mass Number

Consider a nucleus of mass number A:

- Total nuclear mass $M \approx A \times m_u$, where $m_u \approx 1.66 \times 10^{-27}$ kg.
- Nuclear volume:

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi\left(R_0 A^{\frac{1}{3}}\right)^3 = \frac{4}{3}\pi R_0^3 A$$

- Nuclear density:

$$\rho = \frac{M}{V} = \frac{Am_u}{\frac{4}{3}\pi R_0^3 A} = \frac{3m_u}{4\pi R_0^3}$$

Notice that mass number A cancels out completely! Substituting numerical constants:

$$\rho = \frac{3 \times 1.66 \times 10^{-27}}{4 \times \pi \times (1.2 \times 10^{-15})^3} \approx 2.3 \times 10^{17} \text{ kg/m}^3$$

IMPORTANT CONCEPT

Crucial Fact: All nuclei, from lightest hydrogen ($^1_1$H) to heaviest uranium ($^{238}_{92}$U), have the exact same density ($\rho \approx 2.3 \times 10^{17}$ kg/m^3)!

- This colossal density is roughly 2.3×10^{14} times denser than water. A matchbox filled with nuclear matter would weigh over 2 billion tons!

Classification of Nuclear Species

Classification	Condition	Examples
Isotopes	Same atomic number Z, different mass number A (different N). Same chemical properties.	Hydrogen: $^1_1$H, $^2_1$H, $^3_1$H Carbon: $^{12}_6$C, $^{14}_6$C

Classification	Condition	Examples
Isobars	Same mass number A, different atomic number Z. Different chemical properties.	${}^{40}_{18}\text{Ar}, {}^{40}_{19}\text{K}, {}^{40}_{20}\text{Ca}$ ${}^{3}_{1}\text{H}$ and ${}^{3}_{2}\text{He}$
Isotones	Same neutron number $N = A - Z$, different Z and A.	${}^{14}_{6}\text{C}$ $(N = 8)$ and ${}^{16}_{8}\text{O}$ $(N = 8)$ ${}^{3}_{1}\text{H}$ $(N = 2)$ and ${}^{4}_{2}\text{He}$ $(N = 2)$

Fundamental Properties of the Nuclear Strong Force

The nucleons are held together tightly in the tiny nuclear volume against violent Coulomb electrostatic repulsions between protons by the **Nuclear Force**:

1. **Strongest Interaction:** Approximately 100 times stronger than electrostatic Coulomb repulsion and 10^{38} times stronger than gravitational pull at 1 fm.
2. **Short-Range Character:** Operates only over microscopic subatomic distances ($r \leq 2$ to 3 fm). Beyond 3 fm, it drops exponentially to zero.
3. **Charge Independence:** The nuclear strong attraction between two protons (F_{pp}), two neutrons (F_{nn}), and a proton and a neutron (F_{pn}) is identical:
$$F_{pp} = F_{nn} = F_{pn}$$
4. **Saturation Property:** A nucleon interacts only with its immediate nearest neighbors, not with all nucleons in the nucleus. This explains why binding energy per nucleon is nearly constant over a broad mass range.
5. **Repulsive Core at $r < 0.8$ fm**: If nucleons approach closer than 0.8 fm, the force becomes strongly repulsive, preventing the nucleus from collapsing into a mathematical singularity.

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NUCLEAR RADIUS RATIO CALCULATION

Problem: The mass numbers of two nuclei are in the ratio 1 : 27. Find: (a) The ratio of their nuclear radii. (b) The ratio of their nuclear densities.

Solution: (a) From $R = R_0 A^{\frac{1}{3}}$:

$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}} = \left(\frac{1}{27}\right)^{\frac{1}{3}} = \frac{1}{3}$$

(b) Nuclear density ρ is independent of mass number:

$$\frac{\rho_1}{\rho_2} = \frac{1}{1} = 1 : 1$$

Concept Module 5: Mass Defect, Binding Energy Curve & Nuclear Stability

Einstein's Mass-Energy Equivalence

Albert Einstein's special theory of relativity established that mass and energy are interconvertible:

$$E = mc^2$$

- The unified atomic mass unit (1 u) is defined as exactly $\frac{1}{12}$-th the mass of an unexcited carbon-12 ($^{12}_{6}\mathrm{C}$) atom:

$$1\ \mathrm{u} = \frac{1.992647 \times 10^{-26}\ \mathrm{kg}}{12} = 1.660539 \times 10^{-27}\ \mathrm{kg}$$

- Energy equivalent of 1 u:

$$E = (1.660539 \times 10^{-27}\ \mathrm{kg}) \times (2.9979 \times 10^{8}\ \mathrm{m/s})^2 = 1.4924 \times 10^{-10}\ \mathrm{J}$$

$$E = \frac{1.4924 \times 10^{-10}}{1.6022 \times 10^{-13}}\ \mathrm{MeV} = 931.5\ \mathrm{MeV}$$

Mass Defect (Δm)

Accurate mass spectrometry reveals that the rest mass of any stable bound nucleus $M(A, Z)$ is strictly **less** than the sum of the masses of its constituent free protons and neutrons:

$$\Delta m = \left[Z m_p + (A - Z) m_n\right] - M(A, Z)$$

Binding Energy (E_b) and Binding Energy per Nucleon (E_{bn})

The missing mass Δm appears as binding energy holding the nucleons together:

$$E_b = \Delta m c^2 = \Delta m(\text{in u}) \times 931.5\ \mathrm{MeV}$$

The average stability of a nucleus is measured by its **Binding Energy per Nucleon** (E_{bn}):

$$E_{bn} = \frac{E_b}{A}$$

Features of the Binding Energy per Nucleon Curve

A plot of E_{bn} (in MeV/nucleon) against mass number A displays four crucial features:

1. **Global Peak at Iron-56 ($^{56}_{26}\mathbf{Fe}$)**: The curve reaches its absolute maximum at Iron-56 ($^{56}_{26}\mathrm{Fe}$), with $E_{bn} = 8.75$ MeV/nucleon. Iron is the most stable nucleus in nature.
2. **Broad Plateau ($A = 30$ to 170)**: Over this wide intermediate mass range, E_{bn} is nearly constant (≈ 8.5 MeV/nucleon). This plateau is direct proof of the **saturation property** of the nuclear force.
3. **Drop at Heavy Nuclei ($A > 200$)**: For heavy nuclei, cumulative long-range Coulomb repulsion between numerous protons overcomes short-range nuclear attraction. For Uranium-238, E_{bn} drops to ≈ 7.6 MeV/nucleon.
4. **Sharp Spikes at Light Nuclei**: In the light region ($A < 20$), prominent local peaks occur at even-even alpha-like nuclei:
   - Helium-4 ($^{4}_{2}\mathrm{He}$, 7.07 MeV/nucleon).
   - Carbon-12 ($^{12}_{6}\mathrm{C}$, 7.68 MeV/nucleon).
   - Oxygen-16 ($^{16}_{8}\mathrm{O}$, 7.98 MeV/nucleon).

IMPORTANT CONCEPT

Fission & Fusion Energetic Drivers:

- **Fission ($A > 200$):** Heavy nucleus (7.6 MeV/nuc) splits into medium fragments (8.5 MeV/nuc), gaining ≈ 0.9 MeV/nuc, releasing $Q \approx 200$ MeV.

- **Fusion ($A < 20$):** Light nuclei fuse to form ${}^4_2\text{He}$ (7.1 MeV/nuc), gaining $\approx$ 6.0 MeV/nuc, releasing massive energy per unit mass!

BINDING ENERGY OF ALPHA PARTICLE

Problem: Calculate binding energy per nucleon of an alpha particle (${}^4_2\text{He}$). Given: $m_p = 1.00728$ u, $m_n = 1.00867$ u, and $M({}^4_2\text{He}) = 4.00150$ u.

Solution: Mass defect: $\Delta m = [2(1.00728) + 2(1.00867)] - 4.00150 = 4.03190 - 4.00150 = 0.03040$ u. Total binding energy: $E_b = 0.03040 \times 931.5$ MeV $= 28.30$ MeV. Binding energy per nucleon: $E_{bn} = \frac{E_b}{A} = \frac{28.30 \text{ MeV}}{4} = 7.07$ MeV/nucleon.

Concept Module 6: Nuclear Fission, Fusion & Master Revision Cheatsheet

Nuclear Fission: Mechanism & Energetics

Discovered by Otto Hahn and Fritz Strassmann (1938) and explained by Lise Meitner and Otto Frisch:

- A heavy nucleus such as Uranium-235 captures a slow (thermal) neutron ($E_k \approx 0.025$ eV), forming an unstable compound nucleus (${}^{236}_{92}\text{U}^*$), which splits into two medium-sized fragments with release of 2 to 3 prompt fast neutrons:

$$^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{236}_{92}\text{U}^* \rightarrow {}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 3{}^1_0\text{n} + Q$$

- The Q-value is approximately 200 MeV per fission event, distributed primarily as kinetic energy of the fission fragments ($\approx$ 168 MeV), neutrons ($\approx$ 5 MeV), prompt gamma rays ($\approx$ 8 MeV), and subsequent beta-gamma decay ($\approx$ 19 MeV).

Thermonuclear Fusion: Stellar Energy Source

Nuclear fusion occurs when two light nuclei combine to form a single heavier nucleus at extreme temperatures ($T > 10^7$ K) and pressures, overcoming their mutual electrostatic Coulomb barrier. In the core of the Sun, the **Proton-Proton Chain** cycle dominates:

$$4{}^1_1\text{H} \rightarrow {}^4_2\text{He} + 2e^+ + 2\nu_e + 2\gamma + 26.7 \text{ MeV}$$

Energy released per unit mass in fusion is ≈ 6.7 MeV/amu, nearly 8 times higher than fission (0.85 MeV/amu).



ENERGY RELEASED IN NUCLEAR FISSION

Problem: Calculate energy released in fission of 1 g of $^{235}_{92}$U (200 MeV per event, $N_A = 6.023 \times 10^{23}$ mol^{-1}).

Solution: Number of nuclei $N = \left(\frac{1}{235}\right) \times 6.023 \times 10^{23} = 2.563 \times 10^{21}$. Total energy: $E = 2.563 \times 10^{21} \times 200 \times 1.6 \times 10^{-13}$ J $= 8.20 \times 10^{10}$ J ≈ 22.8 MWh (2.5 tons of coal!).

MASTER REVISION FORMULA MATRIX: ATOMS & NUCLEI

- **Distance of Closest Approach:** $r_0 = \frac{2Ze^2}{4\pi\varepsilon_0 K_\alpha} = \frac{4kZe^2}{mv^2}$
- **Bohr Orbital Radius:** $r_n = 0.529\left(\frac{n^2}{Z}\right)$ Å $\propto \frac{n^2}{Z}$
- **Bohr Orbital Speed:** $v_n = \left(\frac{c}{137}\right)\left(\frac{Z}{n}\right) = 2.18 \times 10^6\left(\frac{Z}{n}\right)$ m/s $\propto \frac{Z}{n}$
- **Orbital Energies:** $E_n = -13.6\left(\frac{Z^2}{n^2}\right)$ eV, $K_n = -E_n$, $U_n = 2E_n = -2K_n$
- **Rydberg Transitions:** $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$, $R \approx 1.097 \times 10^7$ m^{-1}
- **Nuclear Radius:** $R = R_0 A^{\frac{1}{3}}$ $(R_0 \approx 1.2$ fm$)$
- **Constant Nuclear Density:** $\rho = \frac{3m_u}{4\pi R_0^3} \approx 2.3 \times 10^{17}$ kg/m^3 (independent of A)
- **Mass Defect & Binding Energy:** $\Delta m = \left[Zm_p + (A-Z)m_n\right] - M$, $E_b = \Delta m \times 931.5$ MeV
- **Fission of U-235:** $Q \approx 200$ MeV; Thermonuclear Fusion: $4\,^1\text{H} \rightarrow {}^4\text{He} + 26.7$ MeV

Practice MCQs & Solutions

Q1. In the Geiger-Marsden alpha-particle scattering experiment, the distance of closest approach for a head-on collision of an alpha particle of kinetic energy K with a gold nucleus of atomic number Z is directly proportional to:

(A) K **(B)** $\frac{1}{K}$
(C) $\sqrt{K}$ **(D)** $\frac{1}{\sqrt{K}}$

Sol. Q1. At closest approach, $K = \left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{2Ze^2}{r_0}\right) \Longrightarrow r_0 = \frac{2Ze^2}{4\pi\varepsilon_0 K} \propto \frac{1}{K}$.

Answer: (B)

Q2. The radius of the first orbit of a hydrogen atom is r_0. The radius of the second orbit of a singly ionized helium ion ($^4_2\mathrm{He}^+$, $Z = 2$) is:

(A) r_0 **(B)** $2r_0$
(C) $4r_0$ **(D)** $\frac{r_0}{2}$

Sol. Q2. Orbital radius scales as $r_n = r_0\left(\frac{n^2}{Z}\right)$. For $^4_2\mathrm{He}^+$, $Z = 2$ and $n = 2$: $r_2 = r_0\left(\frac{2^2}{2}\right) = 2r_0$.

Answer: (B)

Q3. If the total energy of an electron in a hydrogen atom stationary state is -3.4 eV, its kinetic energy and potential energy in this state are respectively:

(A) $+3.4$ eV, -6.8 eV **(B)** -3.4 eV, $+6.8$ eV
(C) $+6.8$ eV, -13.6 eV **(D)** $+3.4$ eV, -3.4 eV

Sol. Q3. For any hydrogenic orbit: $K = -E_n = -(-3.4\text{ eV}) = +3.4$ eV, and $U = 2E_n = 2(-3.4\text{ eV}) = -6.8$ eV.

Answer: (A)

Q4. The ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series of hydrogen is:

(A) $\frac{5}{27}$ **(B)** $\frac{27}{5}$
(C) $\frac{4}{9}$ **(D)** $\frac{9}{4}$

Sol. Q4. Lyman longest ($2 \to 1$): $\frac{1}{\lambda_L} = R\left(1 - \frac{1}{4}\right) = 3\frac{R}{4} \Longrightarrow \lambda_L = \frac{4}{3R}$.
Balmer longest ($3 \to 2$): $\frac{1}{\lambda_B} = R\left(\frac{1}{4} - \frac{1}{9}\right) = 5\frac{R}{36} \Longrightarrow \lambda_B = \frac{36}{5R}$.
Ratio $\frac{\lambda_L}{\lambda_B} = \frac{\frac{4}{3R}}{\frac{36}{5R}} = \frac{4}{3} \times \frac{5}{36} = \frac{20}{108} = \frac{5}{27}$.

Answer: (A)

Q5. An electron in a hydrogen atom de-excites from the $n = 5$ energy level to the ground state ($n = 1$). The maximum number of distinct spectral lines that can be observed is:

(A) 4 **(B)** 8
(C) 10 **(D)** 15

Sol. Q5. Number of spectral lines emitted is $N = \frac{n(n-1)}{2} = \frac{5\times4}{2} = 10$ lines.

Answer: (C)

Q6. The mass numbers of two nuclei are in the ratio 1 : 8. The ratio of their nuclear radii is:

(A) 1 : 2 (B) 1 : 4
(C) 1 : 8 (D) 1 : 1

Sol. Q6. Nuclear radius follows $R = R_0 A^{\frac{1}{3}}$. Therefore, $\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}} = \left(\frac{1}{8}\right)^{\frac{1}{3}} = \frac{1}{2}$.

Answer: (A)

Q7. Two nuclei have mass numbers $A_1 = 27$ and $A_2 = 125$. The ratio of their nuclear mass densities $(\frac{\rho_1}{\rho_2})$ is:

(A) 27 : 125 (B) 3 : 5
(C) 1 : 1 (D) 9 : 25

Sol. Q7. Nuclear density is given by $\rho = \frac{3m_u}{4\pi R_0^3}$, which is completely independent of the mass number A. Hence, the ratio is strictly 1 : 1.

Answer: (C)

Q8. Which of the following pairs represents a pair of isotones?

(A) $^{40}_{18}$Ar and $^{40}_{20}$Ca (B) $^{14}_{6}$C and $^{16}_{8}$O
(C) $^{12}_{6}$C and $^{14}_{6}$C (D) $^{3}_{1}$H and $^{3}_{2}$He

Sol. Q8. Isotones have identical neutron numbers $N = A - Z$. For $^{14}_{6}$C, $N = 14 - 6 = 8$. For $^{16}_{8}$O, $N = 16 - 8 = 8$. Both have 8 neutrons.

Answer: (B)

Q9. The binding energy per nucleon is maximum for which of the following atomic nuclei?

(A) $^{4}_{2}$He (B) $^{12}_{6}$C
(C) $^{56}_{26}$Fe (D) $^{238}_{92}$U

Sol. Q9. The binding energy per nucleon curve reaches its absolute maximum at Iron-56 ($^{56}_{26}$Fe) with $E_{bn} \approx 8.75$ MeV/nucleon, making it the most stable nucleus.

Answer: (C)

Q10. The unified atomic mass unit (1 u) has an energy equivalent of approximately:

(A) 93.15 MeV (B) 931.5 MeV
(C) 9.315 MeV (D) 1.602 MeV

Sol. Q10. By Einstein's mass-energy relation $E = mc^2$: $1\text{ u} = 1.6605 \times 10^{-27}$ kg $\Longrightarrow E = 1.4924 \times 10^{-10}$ J $= 931.5$ MeV.

Answer: (B)

Q11. When a stationary uranium nucleus ($^{235}_{92}$U) undergoes nuclear fission by capturing a thermal neutron, the average energy released per fission is approximately:

(A) 20 MeV (B) 200 MeV
(C) 2000 MeV (D) 2 MeV

Sol. Q11. In the fission of a heavy nucleus like U-235, the binding energy per nucleon increases from ≈ 7.6 MeV to ≈ 8.5 MeV, releasing ≈ 0.9 MeV per nucleon. For 235 nucleons: $Q \approx 235 \times 0.9 \approx 200$ MeV.

Answer: (B)

Q12. In the stellar thermonuclear fusion reaction powering the Sun, four protons fuse together to form a helium-4 nucleus ($^4_2$He). The total energy released in this reaction cycle is approximately:

(A) 26.7 MeV (B) 200 MeV
(C) 13.6 eV (D) 3.27 MeV

Sol. Q12. The overall proton-proton cycle is $4\,^1_1\text{H} \rightarrow\, ^4_2\text{He} + 2e^+ + 2\nu_e + 26.7$ MeV.

Answer: (A)

Q13. The nuclear force between two nucleons is repulsive when the separation distance between them is:

(A) Greater than 3 fm (B) Between 1 fm and 2 fm
(C) Less than 0.8 fm (D) Equal to 1.2 fm

Sol. Q13. The nuclear potential shows a strong repulsive core below $r_0 \approx 0.8$ fm, which prevents the nucleons from collapsing into each other.

Answer: (C)

Q14. The ratio of the speed of an electron in the ground state of a hydrogen atom ($Z = 1, n = 1$) to the speed of light in vacuum (c) is approximately:

(A) $\frac{1}{137}$ (B) $\frac{1}{2}$
(C) $\frac{1}{1836}$ (D) $\frac{1}{73}$

Sol. Q14. In Bohr's theory, ground state velocity is $v_1 = \frac{e^2}{2\varepsilon_0 h} = \frac{c}{137}$. The ratio $\frac{v_1}{c} = \frac{1}{137}$, which is the fine-structure constant.

Answer: (A)

Q15. In a nuclear reaction $A + B \rightarrow C + D$, the total rest mass of the reactants is M_1 and that of the products is M_2. The reaction will release energy (exothermic) only if:

(A) $M_1 = M_2$ (B) $M_1 > M_2$
(C) $M_1 < M_2$ (D) $M_1 = 0$

Sol. Q15. The Q-value is $Q = (M_1 - M_2)c^2$. For an exothermic reaction ($Q > 0$), the initial rest mass must exceed the final rest mass ($M_1 > M_2$), with the mass difference converted into kinetic energy and radiation.

Answer: (B)